Introduction to (Statistical) Machine Learning
Problem Set Five

Summer 2026, IIM B

Due by 11:59 pm on August 21, 2026.

Instructions. All five problems are required. For theoretical problems, using LLMs is not allowed. You are free to use them for the coding problems.

1. Suppose $x_1 = 44, x_2 = 43, x_3 = 47.5, x_4 = 46.5, x_5 = 45$ are observations from the Cauchy distribution with unknown center θ with density

$$\frac{1}{\pi\left(1+(x-\theta)^2\right)} \qquad \text{for all } -\infty < x < \infty.$$

   Assume that the prior distribution for θ is uniform on $[0, 100]$.

   a) Using a fine grid in $[0, 100]$, numerically compute the posterior of θ.

   b) Use the Metropolis–Hastings algorithm to generate a large number of samples from the posterior. Clearly describe which proposal generating density you are using. Tune any hyperparameters of the proposal generating density appropriately to ensure adequate mixing of the Markov chain. Plot the histogram of the samples, and compare it with the numerical evaluation of the posterior density from part (a).

   c) Use PyMC to obtain samples from the posterior. Plot the histogram of the samples. Compare with the histogram of samples from your Metropolis–Hastings chain in part (b), as well as with the numerical evaluation of the posterior density in part (a).

2. The numbers of annual fatal accidents on scheduled airline flights in each of the years 1976–1985 in a specific country are

$$24, 25, 31, 31, 22, 21, 26, 20, 16, 22.$$

   Assume that the number of accidents in year t follows a Poisson distribution with mean $\exp(\alpha + \beta t)$, and assume independence across years. Use a uniform prior for (α, β) over a large set, for example $[-M, M] \times [-M, M]$ for a large M.

   a) Calculate the Laplace posterior normal approximation.

   b) Compute the full posterior by a Metropolis–Hastings algorithm. What is the value of M that you are using? Explain clearly your choice of the proposal generating distribution. Check the trace plots of the samples and comment on whether the Markov chain has mixed adequately. Compare the full posterior to the normal approximation of the previous part, for example in terms of the means and standard deviations corresponding to each parameter.

c) Use PyMC to obtain posterior samples. Check whether the PyMC Markov chain has mixed adequately by looking at the trace plots. Compare the results from PyMC to your results from the previous part.

d) Using each of the three posteriors calculated above (normal approximation, your Metropolis–Hastings sampler, and PyMC), create simulation draws and plot a histogram of the posterior predictive distribution for the number of fatal accidents in 1986. Compute a 95% posterior predictive credible interval for the number of fatal accidents in 1986.

3. Consider the following data on the math scores for a sample of 10th grade students from two US public high schools:

School One (31 students): 52.11, 57.65, 66.44, 44.68, 40.57, 35.04, 50.71, 66.17, 39.43, 46.17, 58.76, 47.97, 39.18, 64.63, 69.38, 32.38, 29.98, 59.32, 43.04, 57.83, 46.07, 47.74, 48.66, 40.80, 66.32, 53.70, 52.42, 71.38, 59.66, 47.52, 39.51.

School Two (22 students): 57.57, 42.40, 41.41, 55.22, 43.90, 53.04, 49.00, 62.45, 53.78, 49.08, 40.25, 43.08, 52.43, 21.73, 53.68, 41.45, 45.47, 34.06, 33.45, 60.78, 35.92, 52.40.

To this data, fit the model

$$
\begin{aligned}
Y_{i1} &= \mu + \delta + \epsilon_{i1}, \\
Y_{i2} &= \mu - \delta + \epsilon_{i2}, \\
\{\epsilon_{ij}\} &\overset{\text{i.i.d}}{\sim} N(0, \sigma^2),
\end{aligned}
$$

with parameters μ, δ, σ^2 along with the prior

μ, δ, σ independent with $\mu \sim N(50, 625)$, $\quad \delta \sim N(0, 625)$, $\quad \sigma^{-2} \sim \text{Gamma}(0.5, 50)$.

a) Use the Gibbs sampler to draw posterior samples for the parameters μ, δ, σ. Implement the Gibbs sampler and report posterior means and standard deviations of μ, δ, σ.

b) Use PyMC to fit this model and compare the results of your Gibbs sampler with those obtained from PyMC in terms of posterior means and standard deviations of μ, δ, σ.

c) Use your posterior samples to calculate the posterior probability that $\delta > 0$.

d) Use your posterior samples to calculate the posterior probability that a randomly sampled tenth grader from the first school will have a higher math score than one sampled from the second school.

4. Consider the Metropolis–Hastings chain which proposes according to a proposal transition kernel $Q(x, y)$ and then accepts or rejects the proposal with acceptance probability

$$
\alpha(x, y) = \min\left(1, \frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)}\right), \tag{1}
$$

where $\pi(\cdot)$ is the target pmf or pdf.

a) Suppose we replace the acceptance probability (1) by

$$
\beta(x, y) = \frac{\pi(y)Q(y, x)}{\pi(y)Q(y, x) + \pi(x)Q(x, y)}.
$$

Show that the resulting Markov chain still satisfies detailed balance with respect to π.

b) More generally, suppose $a : (0, \infty) \to [0, 1]$ is any function satisfying $a(0) = 0$ and $a(u) = u\, a(1/u)$. Show that if the Metropolis–Hastings acceptance probability (1) is replaced by

$$\gamma(x, y) = a\left(\frac{\pi(y)Q(y, x)}{\pi(x)Q(x, y)}\right),$$

then the resulting chain satisfies detailed balance with respect to π.

c) Consider the target distribution $\pi = N(0, 1)$ and the symmetric random-walk proposal

$$Y = X + \sigma Z, \qquad Z \sim N(0, 1),$$

with $\sigma = 1$.

Simulate two Markov chains of length $N = 10,000$: one using the usual Metropolis–Hastings acceptance probability $\alpha(x, y)$, and one using the alternative acceptance probability $\beta(x, y)$ from part (a).

For each chain, report the empirical acceptance rate, sample mean and sample variance, lag-1 autocorrelation, ESS, and a trace plot. Briefly compare the performance of the two chains. Which chain appears to mix better?

5. We briefly discussed the EM (Expectation–Maximization) algorithm in class and its connections to the Gibbs sampler. Consider the problem of computing the MLE by maximizing $\log f_{Y|\theta}(y)$ in a problem with observed data y and parameters θ. Let X be a latent variable and suppose the model can be more naturally written via the full density $f_{Y,X|\theta}(y, x)$. The EM algorithm takes the following steps:

a) Initialize $\theta = \theta^{(0)}$.

b) For $t = 0, 1, 2, \ldots$, repeat:

i. **E-step**: Calculate

$$E(\theta, \theta^{(t)}) = \int \left[\log f_{Y,X|\theta}(y, x)\right] f_{X|Y=y,\theta=\theta^{(t)}}(x)\, dx.$$

ii. **M-step**: Take $\theta^{(t+1)}$ to be the maximizer of $E(\theta, \theta^{(t)})$ over θ.

This problem describes one way of understanding the logic behind the EM algorithm. EM can be seen as a special case of a more general algorithm called MM (Minorize–Maximize or Majorize–Minimize, depending on whether we are solving a maximization or minimization problem).

a) Consider the problem of maximizing a function $F(\theta)$. The MM update from $\theta^{(t)}$ to $\theta^{(t+1)}$ has two steps:

i. Construct a function $G(\theta, \theta^{(t)})$ which minorizes $F(\theta)$ for every θ and agrees with F at $\theta = \theta^{(t)}$:

$$G(\theta, \theta^{(t)}) \le F(\theta) \quad \text{for every } \theta, \qquad G(\theta^{(t)}, \theta^{(t)}) = F(\theta^{(t)}). \tag{2}$$

ii. Take $\theta^{(t+1)}$ to be the maximizer of $G(\theta, \theta^{(t)})$ over θ.

Prove that $F(\theta^{(t+1)}) \geq F(\theta^{(t)})$ for every t.

b) As an example of the MM algorithm, consider maximizing $F(\theta) = \cos\theta$.

   i. Prove that

$$G(\theta, \theta^{(t)}) = \cos\theta^{(t)} - \sin\theta^{(t)}(\theta - \theta^{(t)}) - \frac{1}{2}(\theta - \theta^{(t)})^2$$

   satisfies (2).

   ii. Show that the MM iterate is

$$\theta^{(t+1)} = \theta^{(t)} - \sin\theta^{(t)}. \tag{3}$$

   iii. Prove that the iterative scheme (3) converges to the true maximizer 0 of $F(\theta) = \cos\theta$ when initialized anywhere in the open interval $(-\pi, \pi)$.

c) Recall that the Kullback–Leibler divergence is

$$D(p \mid q) = \int p(x) \log \frac{p(x)}{q(x)}\, dx.$$

Show that $D(p \mid q)$ is always nonnegative, and that $D(p \mid q) = 0$ if and only if $p = q$. **Hint**: use the inequality $u \log u \geq u - 1$, with equality if and only if $u = 1$.

d) With $F(\theta) = \log f_{Y|\theta}(y)$, define

$$G(\theta, \theta^{(t)}) = F(\theta) - D\left(f_{X|Y=y,\theta^{(t)}} \,\middle\|\, f_{X|Y=y,\theta}\right).$$

Show that the conditions (2) are satisfied.

e) Show that $G(\theta, \theta^{(t)})$ has the alternative expression

$$\int f_{X|Y=y,\theta^{(t)}}(x) \log f_{Y,X|\theta}(y, x)\, dx - \int f_{X|Y=y,\theta^{(t)}}(x) \log f_{X|Y=y,\theta^{(t)}}(x)\, dx.$$

f) Show that the MM algorithm applied to $F(\theta) = \log f_{Y|\theta}(y)$ and the above $G(\theta, \theta^{(t)})$ coincides with the EM algorithm.

g) Suppose $\theta = (w, \theta_0, \theta_1)$, $X = (X_1, \ldots, X_n)$, and $Y = (Y_1, \ldots, Y_n)$, with

$$X_i \overset{\text{i.i.d}}{\sim} \text{Bernoulli}(w),$$

and

$$Y_i \mid X_i = 1 \sim N(\theta_1, 1), \qquad Y_i \mid X_i = 0 \sim N(\theta_0, 1).$$

Derive the EM algorithm iterates explicitly. Note that X is discrete here, so $f_{X|Y=y,\theta=\theta^{(t)}}$ should be interpreted as the pmf of X rather than a pdf.